

Jaka Brajadenta

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Motivated Physics graduate from Universitas Pendidikan Indonesia with five years of experience in software development, primarily as a Java Backend Developer using Spring Boot. Adept at building scalable systems, resolving production issues, and developing APIs across digital payment ecosystems. Seeking to contribute robust problem-solving and full-stack development skills in a dynamic technology-driven environment.

PROFESSIONAL EXPERIENCE

FULL STACK PROGRAMMER / PT Jalin Nusantara Pembayaran, South Jakarta in Indonesia
(03/2023) - Present

- Analyzed technical documents (FSD and technical specifications) from System Analysts and prepared development plans based on requirements.
- Developed system programs and modules by applying clean code principles, secure coding practices, and ensuring efficiency according to company standards.
- Conducted unit testing and supported SIT/UAT by preparing test scenarios, collaborating with QA and SA to resolve identified defects.
- Integrated data and systems through APIs, middleware, and digital switching connectors while ensuring transaction performance, reliability, and traceability.
- Managed error logs, exception handling, and debugging to ensure stable solutions in both testing and production environments.
- Supported system deployment to production by preparing technical documentation, performing pre-live validation, and enabling rollback processes when required.
- Debugged and enhanced production-level applications based on incident reports.
- Documented all systems and development processes for internal knowledge transfer, and application administration documents to deliver to production-level.

Project Development

- Payment Integrator as Biller (API Management Synxchro – Javascript ES5)
 - Developed RESTful APIs to connect internal applications with multiple third-party billers.
 - Implemented dynamic routing and fee logic via API gateway (JavaScript/APIM).
 - Built a flexible mapping framework to standardize requests/responses across diverse biller specifications.

- Delivered secure, stable integrations with full technical documentation and SIT/UAT validation.
- QR Domestic (QRIS - MPM) Message Mapper (API Management Synxchro – Javascript ES6)
 - Built an API-based message mapper to unify different QRIS specs into a single standardized API.
 - Implemented dynamic field mapping, caching configurations, and simplified logic with Synxchro APIM new version (JavaScript ES6 + Node.js).
 - Developed secure transaction flows with token-based authentication and HMAC signature validation for QRIS MPM.
 - Enhanced scalability and reduced dependency on third-party switching, increasing business margin.
- Kartu Kredit Indonesia (KKI) Online Payment – CPTS (Java Springboot)
 - Built a core transaction system with Java Spring Boot, enabling secure online KKI payment flows between merchants and issuer banks.
 - Developed REST APIs (token-based) and ISO 8583 TCP interfaces for inter-member and inter-company transactions, with Kafka for data synchronization.
 - Implemented fraud detection checks via external service integration, plus merchant registry validation to prevent unregistered acquirer fraud.
 - Applied dynamic cache system, TLV field mapping, and custom response handling for scalable and compliant national payment operations.
 - Designed a structured logging system for error tracking and system monitoring.
 - Built a backend API for monitoring dashboard, enabling transaction visibility and CRUD operations for configurations (members, routing, products, parameters).
- Near Field Communication (NFC) Payment – based on QRIS message (Java Springboot)
 - Designed a reactive payment system with Java Spring Boot WebFlux, enabling both REST (JSON) and TCP (ISO 8583) gateways for transaction processing.
 - Integrated Kafka for data synchronization, Hazelcast distributed cache, and scheduler-based TCP session management for stable connectivity.
 - Implemented strict security validation, including JWT (with custom algorithms), signature verification, and fraud detection service (FDS) checks.
 - Applied merchant registry validation, dynamic caching, and MySQL persistence to ensure secure, reliable, and compliant NFC-based QRIS transactions.
 - Developed structured logging for easier debugging and monitoring of transaction errors.
 - Built a backend API for dashboard, enabling monitoring, routing management, and CRUD operations for members, products, and app parameters.

JUNIOR APPLICATION DEVELOPER / PT Mitra Integrasi Informatika, West Jakarta in Indonesia (03/2021) - (03/2023)

- Completed 2-month full-stack bootcamp with Java & Spring Boot.
- Developed secure RESTful web applications using Spring Security.
- Produced clean, well-documented code and API specifications.

Project Development

- User Management System (PT Mitra Integrasi Informatika – Java Springboot + Thymeleaf)
 - Developed a basic CRUD web application with Spring Boot for backend (server-side) and Thymeleaf for frontend (client-side).
 - Implemented token-based login and session management to secure user access.
 - Designed user account management features (create, update, delete, and role assignment).
 - Used MySQL database for data persistence and applied structured documentation for training purposes.
- Leave Management System (PT Mitra Integrasi Informatika – Java Springboot + Thymeleaf)
 - Built an advanced leave management platform as bootcamp final project with Spring Boot (backend) and Thymeleaf (frontend).
 - Implemented complex validation logic, including leave date eligibility rules and remaining leave balance calculations.
 - Designed both backend and frontend validation layers for consistent and accurate results.
 - Integrated token-based authentication and session storage for secure login and transaction flow.
 - Utilized MySQL as database with clean CRUD design, achieving a high evaluation score at bootcamp graduation.

On-Site as Outsourcing Employee – JAVA DEVELOPER – PT Jalin Pembayaran Nusantara, South Jakarta in Indonesia (09/2022) - (03/2023)

- Developed claim management websites using Apache Tapestry (Java).
- Automated data parsing from .txt files for reconciliation reports.
- Created financial report generators in .txt and .xlsx formats.

Project Development

- Web Claim Online (Jalin – Java Tapestry)
 - Developed a claim management dashboard using Apache Tapestry for handling disputed transactions in reports.
 - Built role-based access features, enabling different approval workflows for bank members, external operators, and internal admins.
 - Implemented transaction filtering and listing logic, where visibility depended on the logged-in user's role (e.g., member bank vs. internal admin).
 - Added functionality to upload and attach supporting evidence files, allowing reviewers to validate claims with complete documentation.
- Import and Export Dispute Claim (Jalin – Java Spring)
 - Developed a data integration service using Java Spring Native to import dispute data from .txt files into the database for dashboard usage.
 - Implemented export functionality to generate dispute reports from the database into standardized file formats according to industry specifications.

- Ensured structured file formatting (size, layout, encoding) for easy parsing and processing by member banks or principals.
- Improved operational efficiency by automating repetitive data handling tasks for dispute management.
- Reporting Claim and Transaction (Jalin – Java Spring)
 - Built a report generation service with Java Spring Native to produce reconciliation, settlement, and recapitulation reports from Oracle database.
 - Automated daily batch processes to handle large volumes of transaction data and generate multiple reports efficiently.
 - Exported reports into .txt and .xlsx formats following strict specifications for delivery to member banks, participants, and principals.
 - Optimized the application to remain lightweight while ensuring stability and accuracy in high-frequency report generation.

On-Site as Outsourcing Employee – SOFTWARE ENGINEER – PT Astra Digital Artha (AstraPay), South Jakarta in Indonesia (07/2021) - (08/2022)

- Migrated monolith codebases to microservices architecture.
- Led backend development of Cash-in services using Java Spring Boot.
- Integrated top-up, withdrawal, and transfer features via API with partners (banks & aggregators).
- Delivered internal admin and mobile APIs for transaction services.
- Gave knowledge-sharing sessions on AOP and exception handling in Spring Boot.

Project Development

- Cash-In Service Microservice (AstraPay – Java Springboot)
 - Built a microservice API gateway integrated with multiple banks and merchants, enabling users to top up from various channels.
 - Implemented dynamic Bean Validation to minimize redundant queries and ensure efficient request validation.
 - Provided APIs for dashboard monitoring of cash-in transactions.
 - Designed asynchronous batching system to send transaction records to other services for data reconciliation.
 - Supported both REST (JSON/XML) and SOAP web services for third-party integration.
- Cash-Out Service Microservice (AstraPay – Java Springboot)
 - Developed a microservice to handle fund withdrawal and transfers from AstraPay to bank accounts and ATMs.
 - Integrated with third-party APIs as a client, ensuring secure and compliant transaction flows.
 - Applied Bean Validation and service-to-service checks (User Service, Payment Service) for transaction consistency.
 - Implemented batching system for recording and synchronizing transaction logs across services.

- Payment Service Microservice (AstraPay – Java Springboot)
 - Designed a centralized transaction recording service for all financial operations (Cash-In, Cash-Out, Biller, QRIS, etc).
 - Maintained user balance history through debit/credit calculations with automated monthly schedulers.
 - Exposed APIs to provide real-time user balance and transaction history.
 - Served as the backbone for reconciliation and settlement reporting.
- Transaction Service Microservice (AstraPay – Java Springboot)
 - Maintained a legacy transaction recording system, supporting gradual migration to the newer Payment Service.
 - Managed transaction persistence with a different database design, ensuring backward compatibility during migration.
 - Continued to support REST-based integrations while aligning features with the Payment Service.

RESEARCH ASSISTANT / Indonesian Institute of Sciences (LIPI), Bandung in Indonesia
(01/2020) - (02/2020)

- Measured and analyzed impedance characteristics of PZT-8 transducers
- Developed circuits for frequency measurement and reported results to professional researchers

Key Project

- Characteristic Impedance Frequency Transducer Ultrasonic (Experiment and Measurement)

EDUCATION

BACHELOR OF SCIENCE (MATERIAL PHYSICS) / Universitas Pendidikan Indonesia, Bandung in Indonesia and **Graduation Year (2020)** – GPA score 3.54

- Active in Physics Student Association, public events, and academic organizations
- Laboratory Assistant in Fundamental Physics
- Bachelor thesis: Cut-Off Frequency Modeling of TFET on Bilayer Armchair Graphene Nanoribbon using Transfer Matrix Method

ORGANIZATIONS AND AWARDS/ACHIEVEMENTS

- Patent (Simulation Programs): [EC00202297759](#) – Dec 1, 2022

- Patent (Simulation Programs): EC00202138976 – Aug 16, 2021
- Physics Department Community Service (2018 - 2020) - *Committee and Instructure*
- Himpunan Mahasiswa Fisika (HMF) FPMIPA UPI (2017 - 2019) - *Social-Politic Division*
- House of Physics English (HOPE) (2016 - 2017) - *Committee as Public Relations*
- *Scholarship of PPA – Kemenristekdikti. (2018 - 2019)*

WORKSHOP AND TRAININGS

- Secure Code, Multimatics (09/2024)
- Software Test API using Wiremock, Trengganis (08/2024)
- Grafana, Jalin (12/2023)
- PT Mitra Integrasi Informatika, Metrodata Coding Camp – Java (03/2021 – 05/2021)
- ASPEN EC – Council, Certified Secure Computer User (CSCU) (06/2020)

KEYWORDS

Java, Spring Boot, Backend Development, JavaScript, NodeJS, Go, gRPC, REST API, JSON, PostgreSQL, Oracle, MySQL, Postman, Hibernate, ORM, Junit, JIRA, Confluence, Agile, Git, IntelliJ, GoLand, Microservices, Maven Repository, Transaction, Payment, Biller, Financial Technology, Kafka, Redis, CI/CD

REFERENCE

Agus Sentosa (0821-1188-3378) as Former Direct Manager / Jalin Head of Digital Switching Solution